TÉCNICO
UNIVERSIDADE
DE LISBOA

WEEK 1 · DAY 2 · AFTER CLASS

Consolidate Day 2

Extended reading and optional homework on the course site
Not for evaluation

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WEEK 1 · DAY 2 · AFTER CLASS

Extended Reading and Homework for Day 2

Use these to consolidate today. The reading goes deeper than the slides; the homework is optional; none of it is for evaluation. Quiz 1 on the site checks week 1.

W1.3 · Reading 1.4

Sizing by Ratios

From the Arsenal of Venice to Galileo: why ratios from similar ships hold only within a range of size

Open Reading 1.4

W1.3 · Homework H0, optional

Reference Ship Database

Exercise 1.4 in code, a container ship from its TEU, and your first regression of similar ships

Open H0 in Colab

W1.4 · Reading 1.5

Fixed Point and Newton–Raphson

Where x = g(x) and the Newton step come from: the two methods of Lecture 1.5, derived step by step

Open Reading 1.5

W1.4 · Homework H1, optional

Displacement Equation

Solve weight = buoyancy in code: a loop, Newton–Raphson, and a container ship with T last

Open H1 in Colab

Course site

l-u-i-s-b-a-t-i-s-t-a.github.io/PNSO-2026 · all materials, session by session · office hours at calendly.com/vnsd-luisbatista/meetme